

SCQ (Single Correct Type) :

1. If f is a positive function such that $f(x+T) = f(x)$ ($T > 0$), $\forall x \in \mathbb{R}$, then

$$\lim_{n \rightarrow \infty} n \left(\frac{f(x+T) + 2f(x+2T) + \dots + nf(x+nT)}{f(x+T) + 4f(x+4T) + \dots + n^2f(x+n^2T)} \right) =$$

- (A) 2 (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) None of these

2. Let $f(x) = \lim_{n \rightarrow \infty} \tan^{-1} \left(4n^2 \left(1 - \cos \frac{x}{n} \right) \right)$ and $g(x) = \lim_{n \rightarrow \infty} \frac{n^2}{2} \ln \cos \left(\frac{2x}{n} \right)$ then $\lim_{x \rightarrow 0} \frac{e^{-2g(x)} - e^{f(x)}}{x^6}$ equals.

- (A) $\frac{8}{3}$ (B) $\frac{7}{3}$ (C) $\frac{5}{3}$ (D) $\frac{2}{3}$

3. If $g(x)$ is twice differentiable real valued function satisfying $g''(x) - 3g'(x) > 3 \quad \forall x > 0$ and $g'(0) = -1$, then $h(x) = g(x) + x \quad \forall x > 0$ is :

- (A) Strictly increasing (B) Strictly decreasing
(C) Non monotonic (D) Data insufficient

4. If $\int e^x \left(\frac{2 \tan x}{1 + \tan x} + \operatorname{cosec}^2 \left(x + \frac{\pi}{4} \right) \right) dx = e^x \cdot$

$$g(x) + K, \text{ then } g\left(\frac{5\pi}{4}\right) =$$

- (A) 0 (B) 1 (C) -1 (D) 2

5. $I = \int \frac{(\sin 2x)^{1/3} d(\tan^{1/3} x)}{\sin^{2/3} x + \cos^{2/3} x} =$

- (A) $\frac{1}{2^{2/3}} \ln(1 + \tan^{1/3} x) + C$ (B) $\ln(1 + \tan^{2/3} x) + C$
(C) $2^{1/3} \ln(1 + \tan^{2/3} x) + C$ (D) $\frac{1}{2^{2/3}} \ln(1 + \tan^{2/3} x) + C$

MCQ (One or more than one correct) :

6. Let $\lim_{x \rightarrow \infty} (2^x + a^x + e^x)^{1/x} = L$

Which of the following statement (s) is (are) correct?

(A) If $L = a$ ($a > 0$), then the range of a is $[e, \infty)$

(B) If $L = 2e$ ($a > 0$), then the range of a is $\{2e\}$

(C) If $L = e$ ($a > 0$), then the range of a is $(0, e]$

(D) if $L = 2a$ ($a > 1$), then the range of a is $\left(\frac{e}{2}, \infty\right)$

7. Let $f(x) = \frac{\cos^{-1}(1 - \{x\}) \sin^{-1}(1 - \{x\})}{\sqrt{2\{x\}}(1 - \{x\})}$ where $\{x\}$ denotes the fractional part of x , then :

(A) $\lim_{x \rightarrow 0^+} f(x) = \frac{\pi}{4}$

(B) $\lim_{x \rightarrow 0^+} f(x) = \sqrt{2} \lim_{x \rightarrow 0^-} f(x)$

(C) $\lim_{x \rightarrow 0^+} f(x) = \frac{\pi}{4\sqrt{2}}$

(D) $\lim_{x \rightarrow 0^+} f(x) = \frac{\pi}{2\sqrt{2}}$

8. $f(x) = 2e^x + (a^2 - 5a + 6)e^{-x} + (10a - 2a^2 - 11)x - 3$ is increasing for all real values of x if $a \in$

(A) $\{2\}$

(B) $[2, 3]$

(C) $(2, 3)$

(D) $(3, \infty)$

9. Let $f(x) = \int_0^x e^{t^3} (t^2 - 1)t^2(t+1)^{2011}(t-2)^{2012} dt$ at $(x > 0)$ then :

(A) The number of point of inflections is at least 1

(B) The number of point of inflections is 0

(C) The number of point of local maxima is 1

(D) The number of point of local minima is 1

10. There is a group of 6 persons. They play a game in which each has to select a number from 1 to 4. Let A_n is event that n persons have selection of same number, then

(A) $P(A_5) = \frac{18}{4^6}$

(B) $P(A_5) = \frac{18}{4^5}$

(C) $P(A_6) = \frac{1}{4^5}$

(D) $P(A_5/A_6) = \frac{1}{2}$

11. A bag contains four tickets marked with 112, 121, 211, 222 one ticket is drawn at random from the bag. let E_i ($i = 1, 2, 3$) denote the event that i^{th} digit on the ticket is 2. Then :

(A) E_1 and E_2 are independent

(B) E_2 and E_3 are independent

(C) E_3 and E_1 are independent

(D) E_1, E_2, E_3 are independent

12. For two events A and B let, $P(A) = \frac{3}{5}$, $P(B) = \frac{2}{3}$, then which of the following is/are correct ?

(A) $P(A \cap \bar{B}) \leq \frac{1}{3}$

(B) $P(A \cup B) \geq \frac{2}{3}$

(C) $\frac{4}{15} \leq P(A \cap B) \leq \frac{3}{5}$

(D) $\frac{1}{10} \leq P\left(\frac{\bar{A}}{B}\right) \leq \frac{3}{5}$

13. If $\int (\sin 3\theta + \sin \theta) \cos \theta e^{\sin \theta} d\theta = (A \sin^3 \theta + B \cos^2 \theta + C \sin \theta + D \cos \theta + E) e^{\sin \theta} + F$, then:
- (A) $A = -4$ (B) $B = -12$ (C) $C = -20$ (D) None of these
14. If $\int_2^x g(t) dt = \frac{x^2}{2} + \int_x^2 t^2 g(t) dt$, then equation $g(x) = \lambda$ has
- (A) 2 solution if $|\lambda| < \frac{1}{2}$ (B) 2 solution if $|\lambda| < \frac{1}{2}$ & $\lambda \neq 0$
- (C) 1 solution if $\lambda = -\frac{1}{2}$ (D) no solution if $|\lambda| > \frac{1}{2}$

Numerical Based Questions :

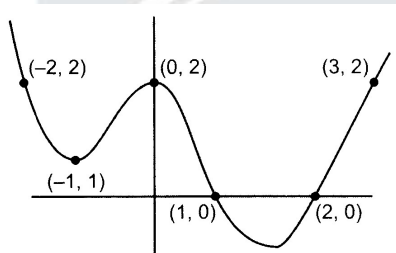
15. $L = \left(\lim_{x \rightarrow 0^+} \frac{6x \{ \ln(\sin x) - \ln x \} + x^3}{(\sin x - x)(1 - \cos x)} \right)$, then $5L$ is equal to.
16. Let $f(x) = \int x^2 \cos^2 x (2x + 6 \tan x - 2x \tan^2 x) dx$ and $f(x)$ passes through the point $(\pi, 0)$, then find the number of solutions of the equation $f(x) = x^3$ in $[0, 2\pi]$

Paragraph Type Question :

Paragraph for Question Nos. 17 to 19

In the given figure graph of:

$y = p(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n$ is given.



17. The product of all imaginary roots of $p(x) = 0$ is :
- (A) -2 (B) -1 (C) $-\frac{1}{2}$ (D) None of these
18. If $p(x) + k = 0$ has 4 distinct real roots $\alpha, \beta, \gamma, \delta$ then $[\alpha] + [\beta] + [\gamma] + [\delta]$, (where $[\cdot]$ denotes greatest integer function) is equal to :
- (A) -1 (B) -2 (C) 0 (D) 1
19. The minimum number of real roots of equation $(p'(x))^2 + p(x) p''(x) = 0$ are :
- (A) 3 (B) 4 (C) 5 (D) 6

Matching Type Problems

20. Let $f(x) = \frac{x^3 - 4}{(x-1)^3} \forall x \neq 1$, $g(x) = \frac{x^4 - 2x^2}{4} \forall x \in \mathbb{R}$, $h(x) = \frac{x^3 + 4}{(x+1)^3} \forall x \neq -1$

	Column-I		Column-II
(A)	The number of possible distinct real equation $f(x) = c$ where $c \geq 4$ can be	(P)	0
(B)	The number of possible distinct real equation $g(x) = c$, where $c \geq 0$ can be	(Q)	1
(C)	The number of possible distinct real equation $h(x) = c$, where $c \geq 1$ can be roots	(R)	2
(D)	The number of possible distinct real roots of equation $g(x) = c$ where $-1 < c < 0$ can be	(S)	3
		(T)	4

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ANSWER KEY OF CAPS – 13+

1. (C) 2. (A) 3. (A) 4. (B) 5. (D) 6. (ABC) 7. (BD)
 8. (ABC) 9. (AD) 10. (BC) 11. (ABC) 12. (ABCD) 13. (ABC) 14. (BCD)
 15. 2 16. 3 17. (D) 18. (A) 19. (B)
 20. $[A \rightarrow Q, R; B \rightarrow R, S; C \rightarrow Q, R, S; D \rightarrow P, R, T]$